Benchmarking Factual Accuracy of Large Language Models Across Scientific Domains

Abstract

Large language models (LLMs) are increasingly used to retrieve, synthesize, explain, and reason over scientific information. Their fluency and broad knowledge make them attractive research assistants, but factual errors, unsupported claims, fabricated citations, and overconfident answers remain important barriers to trustworthy scientific use. Existing evaluation resources are fragmented across general knowledge, biomedical question answering, scientific claim verification, and graduate-level reasoning, making it difficult to compare factual accuracy systematically across scientific domains. This paper reviews the methodological foundations and current approaches for benchmarking LLM factual accuracy across fields including biology, medicine, chemistry, physics, mathematics, and computer science. It distinguishes factual correctness, evidence support, reasoning validity, calibration, and citation faithfulness as related but non-equivalent dimensions of accuracy. Major benchmark families—including MMLU, ARC, GPQA, PubMedQA, SciFact, SciFact-Open, SciQA, and TruthfulQA—are examined in terms of task design, domain coverage, evidence requirements, and limitations. The paper further analyzes automated factuality metrics such as FActScore and SelfCheckGPT, retrieval-augmented generation (RAG), verification-oriented prompting, and automated evaluation frameworks such as ARES. Recent research indicates that external evidence and structured verification can improve reliability, but evaluation itself can be vulnerable to judge-model errors, benchmark contamination, domain imbalance, temporal drift, and differences in question difficulty. The paper argues for a multidimensional, domain-stratified benchmark that combines expert-authored questions, evidence-grounded claim verification, open-ended generation, calibrated uncertainty, citation verification, and temporal robustness. Such a framework would provide a more meaningful assessment of whether LLMs can serve as reliable scientific information systems rather than merely high-performing question-answering models.

Keywords: large language models, factual accuracy, scientific question answering, hallucination, benchmarking, scientific knowledge, factuality, retrieval-augmented generation, claim verification, evaluation

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1. Introduction

Large language models have transformed natural-language processing by demonstrating strong capabilities in question answering, summarization, reasoning, information retrieval, and scientific communication. However, fluency is not equivalent to factual reliability. LLMs can produce statements that are grammatically convincing but unsupported, inaccurate, or inconsistent with established evidence. This problem is particularly consequential in science because scientific answers frequently depend on precise terminology, quantitative relationships, experimental conditions, uncertainty, and distinctions between established findings and emerging hypotheses. Recent surveys identify factuality and hallucination as central reliability problems for LLMs, while emphasizing that factuality evaluation remains difficult for open-ended generation (Wang et al., 2024; Huang et al., 2025).

Scientific factuality is also more demanding than ordinary knowledge recall. A model may correctly identify that a biological mechanism exists but incorrectly state its direction of effect, confuse correlation with causation, misreport an experimental population, or cite a paper that does not support the claim. Consequently, a useful scientific benchmark must evaluate more than whether an answer resembles a reference answer. It should determine whether the claims are correct, whether the reasoning is valid, whether evidence supports the claims, and whether the model appropriately communicates uncertainty.

Several benchmark families have addressed parts of this problem. MMLU evaluates knowledge and reasoning across 57 subjects, including several STEM and professional domains (Hendrycks et al., 2021). ARC evaluates science reasoning using elementary science questions, while GPQA was specifically designed to test difficult graduate-level questions in biology, chemistry, and physics that resist straightforward web search (Clark et al., 2018; Rein et al., 2023). In biomedical science, PubMedQA requires reasoning over research abstracts, while MultiMedQA combines several medical examination, research, and consumer-health datasets (Jin et al., 2019; Singhal et al., 2023).

Other benchmarks directly address evidence-grounded factuality. SciFact introduced scientific claim verification, requiring systems to determine whether research abstracts support or refute scientific claims and to identify relevant rationales (Wadden et al., 2020). SciFact-Open subsequently expanded this problem to a substantially larger corpus and demonstrated that systems developed on smaller collections can suffer significant generalization losses in open-domain scientific evidence retrieval (Wadden et al., 2022).

The central argument of this paper is that no single existing benchmark adequately captures scientific factual accuracy across domains. Instead, a comprehensive evaluation should combine knowledge recall, scientific reasoning, evidence verification, open-ended generation, citation accuracy, uncertainty calibration, and robustness to temporal and domain shifts. The objective is therefore not to propose another isolated question-answering dataset, but to synthesize the requirements for a cross-scientific factuality benchmark.

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2. Background and Related Work

2.1 Factuality and Hallucination in LLMs

Factuality refers broadly to the degree to which generated statements correspond to verifiable information. Hallucination is commonly used for outputs that contain fabricated, unsupported, or contradictory information. Recent work distinguishes factuality hallucination from faithfulness hallucination: the former concerns disagreement with external reality, whereas the latter concerns inconsistency with information supplied in the prompt or context (Huang et al., 2025).

This distinction is important for scientific benchmarking. Suppose a model receives a research abstract and states that a treatment reduced mortality when the abstract reports no significant effect. The response is unfaithful to its source and factually incorrect relative to the scientific evidence. Conversely, a model may produce a scientifically true statement that was not supported by the supplied article. Such a response could be factually correct but fail an evidence-grounding requirement.

Truthfulness also differs from factuality. TruthfulQA demonstrated that models can reproduce common human misconceptions and that increasing model scale does not automatically guarantee truthfulness (Lin et al., 2022). The benchmark contains 817 questions spanning 38 categories and was explicitly designed around questions where plausible but false answers are common. Scientific benchmarks should similarly include misconception-sensitive questions rather than relying exclusively on straightforward textbook facts.

2.2 General Knowledge and Scientific Reasoning Benchmarks

MMLU was designed to measure multitask knowledge and reasoning across 57 subjects. Its coverage includes subjects such as college biology, high-school physics, computer science, medical genetics, and clinical knowledge, making it useful for broad comparisons (Hendrycks et al., 2021). However, its multiple-choice format limits its ability to detect fabricated explanations, unsupported citations, or partial factual errors.

ARC focuses specifically on science questions and contains 7,787 grade-school science questions, divided into Easy and Challenge sets (Clark et al., 2018). Its importance lies in testing scientific reasoning rather than merely language generation, although its educational level does not represent the complexity of contemporary scientific research.

GPQA addresses this limitation through graduate-level questions written by domain experts in biology, physics, and chemistry. The benchmark contains 448 questions, and its authors report that domain experts substantially outperform non-expert validators while frontier models remain challenged by the questions (Rein et al., 2023). GPQA therefore illustrates the value of expert-authored questions for evaluating high-level scientific knowledge.

2.3 Biomedical and Scientific Literature Benchmarks

Biomedical question answering provides one of the most developed areas of scientific LLM evaluation. PubMedQA contains research questions derived from PubMed abstracts and requires yes/no/maybe decisions together with longer answers. Its expert-labeled portion contains approximately 1,000 questions, alongside larger unlabeled and automatically generated sets. The original study reported a substantial gap between model and human performance, demonstrating that biomedical research questions require more than simple lexical matching (Jin et al., 2019).

BioASQ represents another major biomedical evaluation effort, focusing on large-scale biomedical semantic indexing and question answering. Its repeated challenge structure provides an example of how domain-specific evaluation can evolve over time rather than remaining a static benchmark (Nentidis et al., 2022).

Scientific claim verification takes a different approach. SciFact contains approximately 1.4 thousand expert-written scientific claims paired with abstracts containing evidence, with labels indicating whether evidence supports or refutes each claim (Wadden et al., 2020). This format is particularly relevant to factuality because it requires both retrieval and verification.

SciFact-Open extends the task to a corpus of approximately 500,000 research abstracts. Its findings demonstrate a fundamental benchmarking problem: systems can perform well on small, curated collections but lose substantial performance when required to operate over larger, more realistic scientific corpora (Wadden et al., 2022).

2.4 Scientific Knowledge Evaluation

The SciQA benchmark was introduced as a resource for scientific question answering over scholarly knowledge. It illustrates an important direction beyond conventional educational QA: scientific evaluation should increasingly test whether systems can retrieve and reason over scholarly sources rather than simply recall facts from general-purpose training data (Auer et al., 2023).

More generally, human-expert evaluation remains important for scientific factuality. Wysocka et al. (2024) proposed a framework intended to streamline expert evaluation of how accurately LLMs encode scientific knowledge. Their work highlights a fundamental problem: scientific correctness often requires expert judgment because the validity of a claim may depend on specialized terminology, context, methodological details, and interpretation of evidence.

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3. Main Technical Discussion

3.1 What Should "Factual Accuracy" Mean?

A cross-domain benchmark should distinguish at least five dimensions:

1. **Answer correctness:** Is the final answer consistent with the accepted scientific answer?
2. **Atomic factuality:** Are individual factual claims within the response correct?
3. **Evidence support:** Does cited or retrieved evidence actually support the claim?
4. **Reasoning validity:** Does the model correctly infer the answer from premises or scientific evidence?
5. **Calibration:** Does the model express uncertainty when evidence is incomplete or ambiguous?

These dimensions should not be collapsed into a single accuracy number. For example, a model may obtain the correct multiple-choice answer through incorrect reasoning. Conversely, an open-ended response may contain many correct facts but also one serious unsupported conclusion. FActScore addresses the latter problem by decomposing long-form responses into atomic claims and measuring the proportion supported by reliable evidence (Min et al., 2023).

A conceptual benchmark score can therefore be represented as:

$$S_{\text{overall}} = w_1 S_{\text{correctness}} + w_2 S_{\text{evidence}} + w_3 S_{\text{reasoning}} + w_4 S_{\text{calibration}} + w_5 S_{\text{citation}}$$

where the weights should be established in advance rather than optimized retrospectively for a particular model.

3.2 Domain Stratification

Scientific knowledge is heterogeneous. Biology, chemistry, physics, mathematics, medicine, earth science, and computer science use different forms of evidence and reasoning. A benchmark should therefore report results separately by domain rather than producing only an aggregate score.

For example:

Domain	Representative knowledge	Important evaluation capability
Biology	mechanisms, genetics, ecology	conceptual and causal reasoning
Medicine	diagnosis, treatment, evidence	evidence interpretation and uncertainty
Chemistry	reactions, structures, properties	symbolic and quantitative reasoning
Physics	laws, equations, experiments	mathematical and causal reasoning
Mathematics	proofs, computation, definitions	formal reasoning
Computer science	algorithms, systems, complexity	technical and logical reasoning
Earth/environmental science	climate, geology, ecosystems	temporal and empirical reasoning

Such stratification can reveal domain-specific weaknesses that an aggregate score conceals. GPQA already demonstrates the value of expert-authored evaluation in biology, chemistry, and physics, while biomedical benchmarks demonstrate the need for specialized assessment in medicine.

3.3 Benchmark Task Families

A robust cross-scientific benchmark should include several task families.

Closed-book multiple-choice questions test knowledge recall and reasoning. They are inexpensive to score and permit reliable comparison, but they can hide hallucinations because the model is not required to explain its reasoning.

Short-answer questions provide greater freedom but require semantic evaluation.

Open-ended scientific explanations are more realistic but introduce evaluation complexity because answers can be partially correct.

Claim verification asks the model to classify claims as supported, contradicted, or insufficiently supported and to retrieve evidence. SciFact provides an established foundation for this task.

Citation verification evaluates whether a cited paper actually supports the associated statement. This is particularly important because an LLM may generate a real paper title or plausible-looking reference while incorrectly attributing a finding to it.

Abstention tasks ask the model to recognize when evidence is insufficient. This capability is essential in science because "I do not have enough evidence to determine this" can be more scientifically appropriate than an unsupported answer.

3.4 Measuring Atomic Factuality

Long-form answers create a major measurement problem. A response may contain ten correct statements and one incorrect statement, making binary answer-level accuracy inadequate. FActScore addresses this by decomposing responses into atomic facts and calculating the fraction supported by evidence (Min et al., 2023).

For scientific applications, the approach could be adapted to weight claims by importance. A minor terminology error should not necessarily count the same as a false statement about clinical efficacy or experimental results. Thus, a future scientific factuality metric could distinguish:

- major factual errors;
- minor factual errors;
- unsupported but plausible claims;
- citation errors;
- contradictions of evidence;
- appropriately qualified statements.

This would make benchmark results more meaningful for high-stakes applications.

3.5 Evidence-Grounded Evaluation

Evidence-grounded evaluation is especially important in science because many claims are conditional. A statement can be technically correct in one population or experimental setting but misleading when generalized.

SciFact demonstrates how claim verification can connect generated claims to scientific abstracts, while SciFact-Open shows the difficulty of scaling evidence retrieval to a large scientific corpus.

A high-quality benchmark should therefore require models to identify not only a relevant source but also the exact evidence supporting the claim. Evaluation should distinguish:

- **direct support:** evidence explicitly establishes the claim;
- **partial support:** evidence establishes only part of the claim;
- **indirect support:** evidence is related but does not establish the conclusion;
- **contradictory evidence:** the cited source conflicts with the answer;
- **no evidence:** the claim cannot be established from the provided source.

This distinction is crucial because a scientifically relevant citation is not necessarily evidence for the specific statement being made.

4. Current Approaches and Developments

4.1 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) combines parametric language models with external information retrieval. Lewis et al. (2020) showed that retrieval-augmented models can improve performance on knowledge-intensive tasks and generate more specific and factual language than a comparable parametric-only baseline.

For scientific applications, RAG has several advantages. Scientific knowledge changes rapidly, and external retrieval allows systems to access recent papers instead of relying entirely on knowledge encoded during pretraining. Retrieval can also provide provenance and enable users to inspect the evidence.

However, retrieval does not automatically guarantee factuality. A system can retrieve an irrelevant paper, misinterpret the retrieved evidence, or generate a claim that is stronger than what the source supports. Consequently, scientific RAG should be evaluated at three levels: retrieval quality, evidence interpretation, and generation faithfulness.

4.2 Self-Consistency and Hallucination Detection

SelfCheckGPT introduced a black-box approach that samples multiple responses and uses disagreement between samples as a signal of potential hallucination (Manakul et al., 2023). Its underlying intuition is that knowledge-backed statements should be more consistent across samples than unsupported claims.

Although attractive because it does not require access to model probabilities or an external database, self-consistency is not equivalent to truth. A model can consistently repeat the same false statement. Therefore, such approaches should be treated as uncertainty or hallucination signals rather than definitive factuality verification.

4.3 Chain-of-Verification

Chain-of-Verification (CoVe) introduces a structured procedure in which a model drafts an answer, generates verification questions, answers those questions independently, and then revises the original response. Experiments reported reductions in hallucinations across several tasks (Dhuliawala et al., 2023).

For scientific benchmarking, verification-oriented prompting is particularly promising because scientific answers naturally lend themselves to evidence decomposition. A model could be required to identify each major claim, specify what evidence would verify it, retrieve relevant literature, and then revise unsupported claims.

4.4 Automated Evaluation and LLM-as-a-Judge

Automated evaluation has become increasingly important because expert annotation is expensive. ARES, for example, evaluates retrieval-augmented systems along context relevance, answer faithfulness, and answer relevance, using trained lightweight judges together with a relatively small number of human annotations (Saad-Falcon et al., 2024).

However, using an LLM to evaluate another LLM introduces evaluator bias. The judge can share the same misconceptions, fail to recognize subtle scientific errors, or favor fluent answers. This problem becomes especially serious in specialized fields where the evaluator may lack domain expertise.

Therefore, automated evaluation should be calibrated against expert annotations and accompanied by confidence intervals and error analysis rather than treated as an unquestionable ground truth.

4.5 Hallucination Benchmarks

HaluEval provides a broad benchmark for hallucination recognition, containing generated and human-annotated examples across multiple tasks. The authors reported that LLMs can have difficulty recognizing hallucinated information themselves and that external knowledge and additional reasoning can improve recognition.

The implication for scientific benchmarking is important: models should not only be asked to answer questions, but also to critique answers, identify unsupported statements, and determine when evidence is insufficient.

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5. Research Challenges and Limitations

5.1 Benchmark Contamination

One of the largest threats to benchmark validity is training-data contamination. If benchmark questions or their answers appear in model training data, high performance may reflect memorization rather than scientific reasoning. This is particularly difficult for widely used public datasets.

Future benchmarks should therefore use private evaluation sets, newly authored expert questions, dynamic test items, and temporal holdouts where possible.

5.2 Domain Imbalance

Existing benchmarks overrepresent some scientific domains, particularly medicine and general STEM education. Other areas—including specialized chemistry, materials science, astronomy, environmental science, and interdisciplinary research—receive comparatively less systematic coverage.

A cross-domain benchmark should balance not merely the number of questions per field but also the diversity of concepts, difficulty levels, evidence types, and research practices.

5.3 Temporal Knowledge Drift

Scientific knowledge changes. A benchmark based on fixed answers can become outdated, particularly in medicine and rapidly developing fields. A model should not necessarily be penalized for disagreeing with an obsolete consensus.

Future evaluation should record the knowledge cutoff associated with each question and classify questions as temporally stable or temporally dynamic.

5.4 Ambiguity and Scientific Uncertainty

Scientific claims are rarely universally true without qualification. A treatment can be effective for one population but not another. A physical approximation can be valid under particular assumptions. A biological association may not establish causation.

Therefore, benchmark design should reward scientifically appropriate qualification rather than forcing every question into an unconditional true/false format.

5.5 Limitations of Multiple-Choice Evaluation

Multiple-choice benchmarks are attractive because accuracy is easy to calculate, but they do not reveal why a model selected an answer. A model may select the correct option while generating an incorrect explanation.

Conversely, open-ended evaluation is more realistic but harder to score reliably. This creates a trade-off between scalability and scientific validity.

5.6 Human Expert Annotation

Scientific expert annotation is expensive and difficult to scale. Experts may disagree about interpretation, especially in emerging research areas. Nevertheless, expert judgment remains essential for high-quality benchmark construction and validation.

The framework proposed by Wysocka et al. demonstrates the importance of developing more efficient expert evaluation methods for scientific knowledge.

5.7 Citation and Provenance Errors

A model can produce a genuine citation that does not support the associated statement. This creates a particularly dangerous form of hallucination because the presence of a real reference may make an incorrect answer appear trustworthy.

Scientific benchmarks should consequently evaluate citation correctness separately from answer correctness.

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6. Research Gaps and Future Directions

The literature suggests several major gaps.

6.1 A Unified Cross-Domain Benchmark

Existing benchmarks are fragmented. MMLU provides breadth, GPQA provides expert-level scientific difficulty, PubMedQA and BioASQ provide biomedical evaluation, and SciFact provides evidence-based scientific verification.

A future benchmark should combine these complementary properties rather than replacing them with another single task.

6.2 Multidimensional Scoring

Future evaluations should report a profile rather than a single score:

$$\text{Scientific Reliability Profile} = \{\text{knowledge}, \text{reasoning}, \text{factuality}, \text{evidence}, \text{citation}, \text{calibration}\}.$$

This would prevent a model from appearing scientifically reliable merely because it performs well on closed-book multiple-choice questions.

6.3 Expert-Generated Adversarial Questions

Benchmarks should contain questions specifically designed to expose common failure modes:

- plausible misconceptions;
- causal-versus-correlational confusion;
- unit and numerical errors;
- boundary conditions;
- conflicting research findings;
- outdated claims;
- misleading terminology;
- nonexistent or incorrectly attributed studies.

GPQA demonstrates the value of expert-generated difficult questions, while TruthfulQA demonstrates the usefulness of misconception-oriented evaluation.

6.4 Dynamic and Temporal Benchmarking

A long-term benchmark should periodically introduce new questions derived from recently published scientific literature. This would reduce contamination and test whether models can handle newly emerging knowledge.

The benchmark could maintain multiple temporal slices:

- established knowledge;
- recent consensus;
- rapidly changing knowledge;
- unresolved scientific questions.

6.5 Evidence-Centered Scientific RAG Evaluation

RAG systems should not be judged solely on whether they retrieve relevant documents. Future benchmarks should measure the complete chain:

$$\text{Question} \rightarrow \text{Retrieval} \rightarrow \text{Evidence} \rightarrow \text{Reasoning} \rightarrow \text{Claim} \rightarrow \text{Citation.}$$

ARES provides a useful foundation because it separately evaluates context relevance, answer faithfulness, and answer relevance.

6.6 Calibration and Abstention

Scientific reliability requires knowing when not to answer. Future benchmarks should therefore measure whether model confidence corresponds to correctness and whether models appropriately abstain when evidence is insufficient.

This is particularly important because recent work has highlighted an uncomfortable interaction between evaluation objectives and hallucination behavior: optimizing systems around simple accuracy criteria can create incentives for models to answer questions even when they should express uncertainty (Kalai et al., 2026).

6.7 Independent and Reproducible Evaluation

Benchmark results should ideally be reproduced across model versions, inference settings, prompting strategies, and independent evaluation pipelines. This is necessary because LLM behavior can vary with system instructions, sampling parameters, context length, retrieval configuration, and model updates.

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7. Proposed Benchmarking Framework

Based on the reviewed literature, a comprehensive benchmark could be organized into six stages.

Stage 1: Domain-balanced question construction.
Expert researchers create questions covering biology, medicine, chemistry, physics, mathematics, computer science, and other scientific domains. Questions should span foundational, advanced, quantitative, conceptual, and interdisciplinary knowledge.

Stage 2: Independent expert validation.
At least two qualified experts independently verify each question, answer, difficulty level, and supporting sources. Disagreements should be adjudicated rather than hidden.

Stage 3: Multiple evaluation modes.
Each scientific concept should, where feasible, be represented through multiple formats: multiple choice, short answer, open-ended explanation, claim verification, and evidence retrieval.

Stage 4: Evidence and citation assessment.
Models should provide supporting sources for open-ended scientific claims. Evaluators should verify

whether the cited source exists, whether it contains the claimed information, and whether the model accurately represents the source.

Stage 5: Calibration and abstention testing.
Models should assign confidence to their answers and receive credit for appropriately refusing to make unsupported claims.

Stage 6: Error taxonomy and reporting.
Rather than reporting only accuracy, benchmark reports should classify errors into factual errors, reasoning errors, citation errors, evidence errors, numerical errors, temporal errors, and overconfidence.

Such a benchmark would build on the complementary strengths of existing resources while addressing their fragmentation.

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8. Conclusion

Benchmarking factual accuracy in large language models requires more than measuring answer accuracy on a conventional question-answering dataset. Scientific knowledge is specialized, evidence-dependent, uncertain, and continuously changing. Existing benchmarks such as MMLU, ARC, GPQA, PubMedQA, SciFact, SciFact-Open, and SciQA have made important contributions, but each evaluates only part of the broader problem.

The literature increasingly supports multidimensional evaluation involving factual correctness, evidence grounding, citation faithfulness, reasoning, and uncertainty. Techniques such as retrieval-augmented generation, self-consistency, Chain-of-Verification, atomic factuality metrics, and automated RAG evaluation can improve either model reliability or evaluation scalability, but none eliminates the need for expert oversight.

The most promising direction is therefore a domain-stratified, evidence-centered, temporally aware benchmark that combines expert-authored questions with open-ended scientific generation and claim verification. Such an evaluation should reward not only correct answers but also accurate evidence use, appropriate uncertainty, and faithful citation. Ultimately, the goal should not be to identify which LLM produces the most convincing scientific prose, but which systems can provide **verifiable, reproducible, appropriately qualified, and scientifically defensible information**.

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